MASTER'S THESIS
(PHYSICS)

Evolution of Open Quantum Systems under Continuous Measurement of their Environment

STUDENT:
Jules STEVENOT

SUPERVISOR:
Bruno BELLOMO

January-June 2026

Deliberately left blank

ACKNOWLEDGEMENTS

The author expresses deep gratitude to Bruno Bellomo for his guidance and support throughout this research. He is a man who always took the time to explore ideas, ensure our understanding, and seek details. His expertise in Quantum Mechanics has been fundamental to the successful completion of this Master's thesis. I am grateful to conclude my short career as a physicist with such a demanding and rewarding journey, thanks to Bruno.

I would like to thank the members of the jury for their time and interest in this work, as well as for the opportunity to present my work in front of such esteemed peers. I also would like to thank my colleagues for their insightful discussions and moral support during this adventure: Silvio Da Silva, Jana El Badawi, Anjana Mohan, and Léo Jacquerot-Legros for welcoming me into your office. Thank you in particular for your time and the discussions we had, which greatly helped me.

Finally, I would like to thank my friends and family for their never-ending support and love: Erwan Le Dœuff, Titouan Alphonse, and Nabyan Arkan for these two years of Master's studies and friendship together; Maxence Bonin and Vincent Girod for being there like brothers when difficulties arose; my parents Catherine and Lionel for my entire life and education; my brother Pierre for his righteousness and compassion; and finally my other half, without whom I would not be who I am today, Alexiane.

L'auteur exprime sa profonde gratitude à Bruno Bellomo pour ses conseils et son soutien tout au long de cette recherche. C'est quelqu'un qui sait prendre le temps d'explorer des idées, de s'assurer de notre compréhension et de rechercher les détails. Son expertise en mécanique quantique a été fondamentale pour la réussite de ce mémoire de Master. Je suis reconnaissant de conclure mon court parcours de physicien par une aventure aussi exigeante et gratifiante, grâce à Bruno.

Je tiens à remercier les membres du jury pour le temps qu'ils ont consacré à ce travail et pour l'intérêt qu'ils lui ont porté, ainsi que pour l'opportunité de présenter mes travaux devant des pairs reconnus. Je souhaite également remercier mes collègues pour leurs discussions enrichissantes et leur soutien moral au cours de cette aventure : Silvio Da Silva, Jana El Badawi, Anjana Mohan et Léo Jacquerot-Legros pour m'avoir accueilli dans leur bureau. Je vous remercie tout particulièrement pour votre temps et pour les discussions que nous avons eues, qui m'ont grandement aidé.

Enfin, je tiens à remercier mes amis et ma famille pour leur soutien et leur amour indéfectibles : Erwan Le Dœuff, Titouan Alphonse et Nabyan Arkan pour ces deux années d'études de Master et notre amitié partagée ; Maxence Bonin et Vincent Girod pour avoir été comme des frères lorsque des difficultés sont survenues ; mes parents, Catherine et Lionel, pour toute ma vie et mon éducation ; mon frère Pierre pour sa droiture et sa compassion ; et enfin ma moitié, sans qui je ne serais pas qui je suis aujourd'hui, Alexiane.

« La gloire de Dieu, c'est de cacher les choses; La gloire des rois, c'est de sonder les choses. »
— **Proverbes 25:2**

CONTENTS

Introduction **1**

I From closed systems to continuous measurements **2**

I.i) Quantum mechanics 2

I.i.1⟩ States and time evolution 2

I.i.2⟩ Observables and purity of states 2

I.i.3⟩ Postulates of quantum mechanics 3

I.ii) Open quantum systems 3

I.iii) Quantum measurement theory 3

I.iv) Stochastic master equation 3

CONTEXT AND MOTIVATIONS

This document is the report of an internship carried out as part of a Master's degree in fundamental and numerical physics (CompuPhys).
My internship has been conducted within the team of Theorical Physics — *équipe de Physique Théorique* ΦTh — of the *UTINAM Institute* in the building of the *Observatory of Besançon*.
It aims to study from an academical perspective at first, the subject of continuous measurement considering open quantum systems. The main objective was to explore a subject almost-unknown in details to either my supervisor or myself. We wanted to understand the possibilities and how to handle this domain properly. This work could then lead to other searches linked to continuous measurements.

By familiarizing ourselves with the topic, we decided to adopt this perticular frame of report. The structure of this document is meant to help any unfamiliar reader (about continuous measurement), to discover and explore its fundamentals. We wanted to make it as illustrative and pedagogical as possible, thus, simple examples will be used to graphically depict mathematical and physical concepts.

Introduction

"Quantum mechanics is the description of the behavior of matter and light in all details and, in particular, of the happenings on an atomic scale" said *R.Feynman* in one of his lectures [1]. In this framework, position is no longer a single well-defined value prior to measurement; instead, it is described by a probability density all across space. Consequently, the notion of a trajectory as a continuous succession of points in spacetime is lost in quantum dynamics. We will see further below how to define these trajectories in the case of our study, but one question remains paramount: how can we even define a quantum system?

In the ideal case, systems are treated as *closed*, *i.e.*, isolated from any external environment and interacting with nothing at all. This modelling allows quantum mechanics to describe dynamics of systems with unitary and deterministic terms (the state at time t is determined solely by knowing the state at time t_0) using the Schrödinger equation. [2] However in the physical world, systems do "feel" their surroundings, *i.e.*, they interact with their environment. This interaction should be taken into account to obtain a more complete and physically relevant representation; in doing so, we describe the so-called *Open Quantum System* (OQS) theory [3].

Useful theories have been developed through the last decades, allowing quantum description to be more and more complete. One of them being the Quantum Measurement Theory (QMT) [4], where measurement is fully described. Measuring a system has a tremendous impact on the quantum world, we will take time to explain why further below. However, it is the very role of any experiment to interact with a system in order to measure it. The inquiry of measuring a system without impacting its quantum properties too much leads to the concept of weak measurements.

By studying QMT, one can thus notice that measuring a quantum system alters it. The measurement usually interacts so strongly with the system that it collapses into a state — an eigenstate of the observable — that lost interesting quantum properties, *e.g.*, losing entanglement[1]. Or, quantum systems can be in superposition of states, taking *Schrödinger's cat* thought experiment: the cat can be alive and dead at the same time before interaction. Measuring the state of the cat will determine its fate, alive or dead, it is then "stuck" and cannot be written as a superposition again. The idea of **weak measurements** is to **measure the environment** coupled with the system. Instead of altering the state, we obtain pieces of information on the system. The measurements can even be conducted continuously, maximizing the amount of data the observer can get. These are the global ideas behind **Quantum Continuous Measurements** (QCM) [5].

This work is organized as follows: first section covers an academical progression from quantum closed systems up to the evolution of open quantum systems under continuous photodetection/homodyne detection; following the qubit and harmonic oscillator examples. The second part is a study of a system of interest on which we apply quantum continuous measurement. (Si on a le temps.)

[1]For example, $|\psi\rangle_{AB} = \frac{|00\rangle+|11\rangle}{\sqrt{2}} \xrightarrow{P_0} |00\rangle_{AB} = |0\rangle_A \otimes |0\rangle_B$ a product state (thus non-entangled).

I From closed systems to continuous measurements — in a nutshell

The examples and illustrations you can find in the following are made on `Julia` using `QuantumToolbox.jl` especially to simulate the dynamics.

I.i) Generalities on quantum mechanics

I.i.1⟩ States and time evolution

In quantum mechanics a physical system's state is described by a vector of a mathematical space, a Hilbert space:

$$|\psi\rangle \in \mathcal{H}$$

The evolution of a system depends on its energy, the Hamiltonian H is an hermitian[2] operator that accounts for the total energy of a system. The Schrödinger equation states it explicitly:

$$i\hbar \frac{\mathrm{d}\,|\psi(t)\rangle}{\mathrm{d}t} = \hat{H}(t)\,|\psi(t)\rangle$$

Instead of looking at the state at time t we can use an unitary[3] operator $\hat{U}$ such that, $|\psi(t)\rangle = \hat{U}(t,t_0)\,|\psi(t_0)\rangle$. From now on, we will omit the $\hat{}$ symbol over operators when it is obvious. We get,

$$\dot{U}(t,t_0) = -\frac{i}{\hbar} H(t) U(t,t_0)$$

One can solve this differential equation in general by using a chronological time-ordering operator [3]. Giving for the state evolution,

$$|\psi(t)\rangle = \mathcal{T}_{\leftarrow} \exp\left[-\frac{i}{\hbar}\int_{t_0}^{t} H(s)\mathrm{d}s\right]|\psi(t_0)\rangle$$

In simpler terms, when the Hamiltonian is independent of time,

$$|\psi(t)\rangle = \underbrace{e^{-\frac{iH(t-t_0)}{\hbar}}}_{U(t,t_0)}|\psi(t_0)\rangle$$

This evolution of the state through time is said **unitary and reversible** because the U is unitary, and, because this equation also describe the time reversed evolution $t \to t_0$: $|\psi(t_0)\rangle = U^\dagger(t,t_0)\,|\psi(t)\rangle$.

I.i.2⟩ Observables and purity of states

Any physical quantity is represented in quantum mechanics by an **observable** which is an operator whose eigenvectors form an orthonormal basis of the Hilbert space $\mathcal{H}$ [2]. For finite dimension Hilbert space, hermitian operators[4] are observables. In the following we will consider finite dimension, non-degenerated cases to illustrate but one can generalize [2].

[2] $\hat{H}^\dagger = \hat{H}$, where † is the conjugate transpose operation, defining the hermitian adjoint.

[3] $UU^\dagger = U^\dagger U = \mathbb{I}$, where $\mathbb{I}$ is the identity.

[4] Hermitian operators always form a basis in finite dimension; we can use Gram-Schmidt process to find its orthonormal counterpart.

A natural operator whose eigenbasis is particularly interesting is the Hamiltonian, its eigenvalues correspond to the possible energy levels of a system.
Let us call $\{\phi_n\}_n$ the family of orthonormal eigenvectors of H which forms a basis of $\mathcal{H}$. Since it forms a basis of the Hilbert space, one can describe any state in this very basis:

$$\forall\, |\psi\rangle \in \mathcal{H}; \qquad |\psi\rangle = \sum_{n=1}^{\dim(\mathcal{H})} \underbrace{c_n}_{\langle\phi_n|\psi\rangle} |\phi_n\rangle$$

This kind of description in a base is obviously valid for any observable, not only with the Hamiltonian.

I.i.3⟩ Postulates of quantum mechanics

Physicists developed a corpus of the main properties of quantum mechanics, here are summarized the so-called postulates of quantum mechanics [2]:

1. The state of a system is represented by a density matrix $\rho \in \mathcal{L}(\mathcal{H})$; for pure states we can describe them using $|\psi\rangle$ (seen in I.i.1⟩), pure density matrix becomes simply $\rho = |\psi\rangle\langle\psi|$.

2. Physical quantities $\mathcal{A}$ are associated with operators $\hat{A}$, called observables. In the context of finite dimension space, they are simply Hermitian operators.

3. The only possible outcomes of a measurement of $\mathcal{A}$ are the eigenvalues a of $\hat{A}$, defined by $\hat{A}\,|a\rangle = a\,|a\rangle$.

4. When measuring $\mathcal{A}$ on a normalized system $|\psi\rangle$, we know the probability to get one of the possible eigenvalues: $\mathbb{P}(a_n)$ (which depends on the spectrum of the observable).

5. If the measurement of the physical quantity $\mathcal{A}$ on the system in the state ψ gives the result a_n, the state of the system immediately after the measurement becomes the eigenvector ϕ_n.

6. The time evolution of the wavefunction $|\psi(t)\rangle$ is governed by the Schrödinger equation:
$$i\hbar\frac{\mathrm{d}\,|\psi(t)\rangle}{\mathrm{d}t} = \hat{H}(t)\,|\psi(t)\rangle$$
where $\hat{H}$ is the hamiltonian of the system.

The kind of dynamics a system follows is then unitary and deterministic. We can always solve, at least analytically, for the solution of the Schrödinger equation.

I.ii) Open Quantum Systems (OQS)

I.iii) Quantum Measurement Theory (QMT)

I.iv) Stochastic Master Equations (SME)

References

[1] Richard P. Feynman, Robert B. Leighton, and Matthew Sands. *The Feynman Lectures on Physics, Vol. III, Chapter 1: Quantum Behavior*. California Institute of Technology, accessed 2026-04-02. 1965. URL: https://www.feynmanlectures.caltech.edu/III_01.html.

[2] Claude Cohen-Tannoudji, Bernard Diu, and Franck Laloë. *Mécanique quantique*. fr. Companyédition CNRS/EDP Sciences, Sept. 2018.

[3] Heinz-Peter Breuer and Francesco Petruccione. *The Theory of Open Quantum Systems*. Oxford University PressOxford, Jan. 2007. ISBN: 9780191706349. DOI: 10.1093/acprof:oso/9780199213900.001.0001. URL: http://dx.doi.org/10.1093/acprof:oso/9780199213900.001.0001.

[4] Kurt Jacobs. *Quantum Measurement Theory and its Applications*. en. Cambridge, England: Cambridge University Press, Aug. 2014.

[5] Francesco Albarelli and Marco G. Genoni. "A pedagogical introduction to continuously monitored quantum systems and measurement-based feedback". In: *Physics Letters A* 494 (Jan. 2024), p. 129260. ISSN: 0375-9601. DOI: 10.1016/j.physleta.2023.129260. URL: http://dx.doi.org/10.1016/j.physleta.2023.129260.